Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	*(a)*	(i)		Mean = 50 cm (1) [accept 49.8 cm] Uncertainty = $\frac{(54 - 46)}{2}$ = [±] 4 cm (1) [accept 4.0 cm] **Alternative for 2nd mark:** Uncertainty = 54 - 49.8 = 4.2 cm (1) [accept 4.0 cm or 4 cm]		2		2	2	2
		(ii)		% uncertainty = $\left(\frac{4}{50}\right)$ x 100 **(ecf on mean and uncertainty)** = ± 8% accept 8.4 % if alternative method used in (i) allow 1 or 2 sig figs		1		1	1	1
	(b)	(i)		Conservation of energy (1) accept energy can't be created or destroyed – don't accept principle of energy Energy held in / work done by compressed spring transferred to gravitational potential energy (1) Accept: ½ $kx^2 = mgh_{\text{mean}}$	1	1		2		
		(ii)	I	Very small uncertainty in both measurements **or** high precision **or** small resolution [when compared to uncertainty in jump height] (1) Accept calculations – 0.3% and 0.02%. Don't accept reference to accuracy Hence negligible effect on [overall] uncertainty **or** when compared to 8% (1)			2	2		2
			II	Substitution: e.g. $k = \frac{(2 \text{ x } 48.4 \text{ x } 10^{-3} \text{ x } 9.81 \text{ x } 50)}{3.20^2}$ (**ecf on mean**) Ignore powers of 10 (1) k = 4.6 N cm^{-1} **or** 460 N m^{-1} **(unit mark)** (1) accept 4.62 or 462 if 49.8 used and unit J m^{-2} Uncertainty = ± 8% × 4.64 = ± 0.4 N cm^{-1} or 40 N m^{-1} (1) (**ecf on both % uncertainty and *k***) **Allow 1 or 2 sig figs or same number of sig figs as *h*** **Accept answers when uncertainties in mass and *x* have been calculated by candidates e.g. total uncertainty = 8.32%**	1	1 1		3	3	3